QUANTITATIVE DISTORTION BOUNDS AND FORMAL EQUILATERAL EXCLUSION FOR ELEVEN POINTS IN ℓ_1^5

ARGUS OPEN RESEARCH PROJECT

Abstract. Kusner's 1983 conjecture asserts that an equilateral set in ℓ_1^d has cardinality at most $2d$. In dimension $d = 5$, we establish two complementary quantitative results on the minimum aspect ratio $\Delta(X) = \max_{i<k} \|x_i - x_k\|_1 / \min_{i<k} \|x_i - x_k\|_1$ over eleven-point configurations $X \subset \ell_1^5$. First, by developing a perturbation-stable version of terminal coordinate clipping, reciprocal-mass weighting, and span majorization, we prove the universal lower bound $\Delta(X) > \frac{3089}{3084} > 1.00162$. Second, by optimizing over coordinate-order cones with exact rational primal-dual linear programming certificates, we construct an explicit eleven-point rational configuration with denominator 848 achieving $\Delta(X) = \frac{115}{106} \approx 1.084906$, strictly improving upon the symmetric centered benchmark $\frac{9}{8} = 1.125$. In addition, we classify the complete 66-graph five-edge endpoint-incidence relaxation cover and record a complete Lean 4.33.1 kernel verification of the exact $e(\ell_1^5) = 10$ exclusion.

1. Main Mathematical Results

Theorem 1.1 (Quantitative eleven-point distortion bounds in ℓ_1^5)**.** *Let* $\Delta_5(11) = \inf\{\max_{1\le i<k\le 11} \|x_i - x_k\|_1 / \min_{1\le i<k\le 11} \|x_i - x_k\|_1 : x_1, \dots, x_{11} \in \mathbb{R}^5$ *distinct*$\}$. *Then*

$$\frac{3089}{3084} < \Delta_5(11) \le \frac{115}{106}.$$

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 7200051 in the domain of discrete and metric geometry adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Strict quantitative lower bound > 3089/3084 and rational upper bound <= 115/106 on the 11-point aspect ratio in l_1^5, plus Lean 4.33.1 kernel audit of e(l_1^5)=10.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Exact value of the unrestricted 11-point minimum aspect ratio in l_1^5 and Kusner's conjecture in dimensions d >= 7

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 7200051 (discrete and metric geometry), archived in the accompanying repository package `results/7200051/`.

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